

METHOD NOTE / 04

Small signals, better decisions

A mathematical note on confidence-weighted prioritization

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1 The decision problem

Product teams regularly compare opportunities with unequal evidence. One item may have a clear customer signal but a difficult implementation; another may be cheap to ship but supported only by intuition. A useful prioritization rule should reward expected impact while making uncertainty **visible** rather than implicit.

Definition — a decision-ready bet

A bet is decision-ready when the team can state its intended impact, its current confidence, its estimated effort, and the next piece of evidence that could change the decision.

The model below is not a substitute for judgment. It is a compact language for making that judgment inspectable. Its output is a ranking aid, not a promise of value.

1.1 A confidence-weighted score

For each candidate i , let I_i represent expected impact, C_i represent confidence, and E_i represent estimated effort. We use the following score:

$$S_i = \frac{I_i \cdot C_i}{E_i} \quad (1)$$

Impact and confidence are expressed on the unit interval. Effort is a positive relative estimate, so the score answers a narrow operational question: **which opportunity currently offers the most plausible impact per unit of attention?**

To keep confidence honest, derive it from three independently discussable inputs:

$$C_i = 0.45u_i + 0.35q_i + 0.20f_i \quad (2)$$

where u_i is user evidence, q_i is solution quality evidence, and f_i is delivery feasibility. The weights can change, but the separation is valuable: it prevents one compelling interview or one implementation preference from standing in for the entire case.

1.2 A worked example

Consider three potential bets for a subscription product. All numbers below are illustrative and normalized for comparison.

Candidate	I	C	E	S
Trial reminder	0.70	0.82	0.45	1.28
Guided setup	0.88	0.62	0.80	0.68
Usage digest	0.56	0.76	0.30	1.42

Table 1: A simple comparison using the confidence-weighted score in Equation 1.

The usage digest ranks first, even though its raw impact estimate is modest. It has enough evidence and sufficiently low effort that it becomes an attractive learning move. That does **not** mean the team should ignore the guided setup; it means the next discussion should ask how to increase the confidence of that higher-impact bet.

2 From score to operating behavior

The score becomes useful when it prompts a specific next action. A low score can indicate weak evidence, high effort, low impact, or some combination of all three. The remedy should depend on the source of the weakness.

High impact, low confidence

Run the smallest test that can change C_i . Do not begin full delivery merely because I_i is attractive.

High confidence, high effort

Break the work into a smaller decision. Look for a reversible first slice that reduces E_i or clarifies the constraint.

2.1 A reference implementation

The rule is intentionally lightweight. A short function is often enough to make the model visible in a planning artifact.

```
from dataclasses import dataclass

@dataclass
class Bet:
    impact: float
    confidence: float
    effort: float

def priority(bet: Bet) -> float:
    return (bet.impact * bet.confidence) / bet.effort
```

The implementation is less important than the review habit it creates. Teams should keep the inputs near the decision, preserve the reason behind a score, and revisit assumptions after the next learning cycle.

2.2 Sensitivity and restraint

A score is useful only if small changes in an uncertain input do not create false certainty. One practical check is to vary confidence by a modest range, δ , and observe whether the ranking changes:

$$\Delta S_i = \frac{I_i(C_i + \delta)}{E_i} - \frac{I_i C_i}{E_i} = \frac{I_i \delta}{E_i} \quad (3)$$

Equation 3 tells us that the score is most sensitive for high-impact, low-effort candidates. That is precisely where teams should document their evidence carefully: a small confidence revision can alter the order of work.

Caution. The model should never be used to “win” a decision. Its purpose is to expose assumptions, make trade-offs discussable, and give new evidence somewhere to land.

3 Conclusion

The useful outcome is not a perfect ranking. It is a planning conversation that distinguishes evidence from ambition, knows which uncertainty is worth reducing, and can make a clear next move. Use Table 1 as a

starting point, then tune the weights only after the team has learned whether they improve the quality of its decisions.